

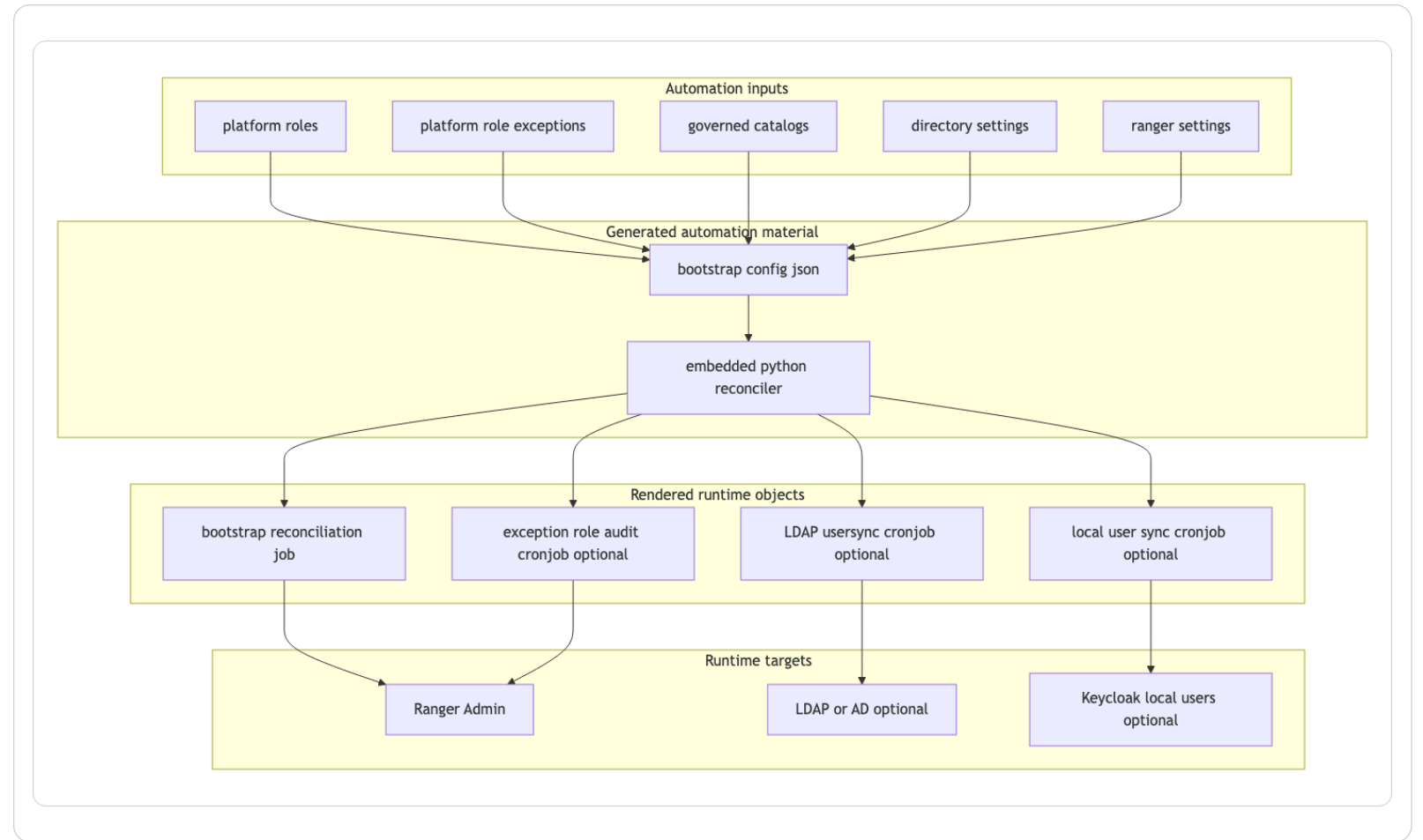


Session 06 / Roles, policies,  
and enforcement

# Governance, Authorization, And Ranger

One hour handover deck

40-45 min narrative / 10-15 min checkpoint /  
5 min Q&A



**Ranger can exist as a governance service even when Trino is still authorized by generated file rules.**

# Session path

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- 1 **Governance concepts**
- 2 **Validation rules**
- 3 **Ranger automation**
- 4 **Trino enforcement matrix**

## Session rhythm

40-45 min narrative  
10-15 min checkpoint  
5 min Q&A

**Presenter notes carry the script and demo prompts.**

# Governance is part of the chart contract

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## Not just infrastructure wiring

- Platform roles describe durable app and data entitlements.
- Direct-user exceptions require approval metadata and expiry.
- Governed catalogs carry PI, IRB, consent, and approval metadata.
- Ranger bootstrap policies can be reconciled into Ranger.

**Platform  
roles  
describe  
durable app  
and data  
entitlements.**

# Core authorization values

| Values path                                   | What it means                                                    |
|-----------------------------------------------|------------------------------------------------------------------|
| global.authorization.platformRoles            | Named roles mapped into app entitlements and Ranger roles        |
| global.authorization.platformRoleExceptions   | Controlled direct-user exceptions with approval metadata         |
| global.dataCatalogs.<name>.governance         | Catalog-level metadata that controls whether data may be exposed |
| global.authorization.ranger.bootstrapPolicies | Policy definitions reconciled into Ranger                        |

# Governance validation prevents unsafe combinations

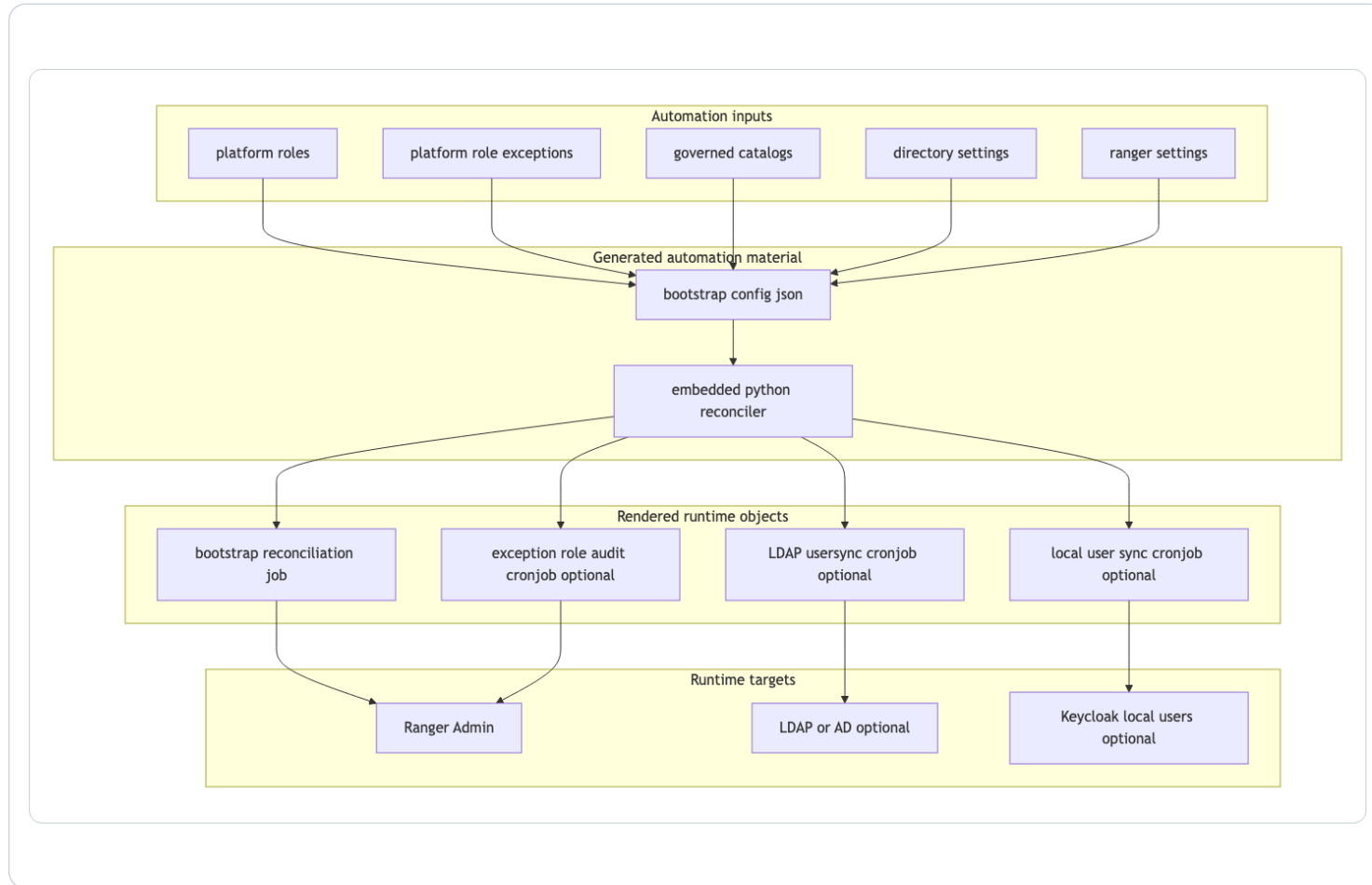
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## Representative enforced rules

- Every platform role needs a description.
- App entitlements must use supported app names.
- Direct-user exceptions need approval reference, reason, grantor, and expiry.
- Restricted data cannot rely on unsafe wildcard access patterns.
- Identifiable restricted data needs masking or row-filter policy coverage.

**Every  
platform role  
needs a  
description.**

# Ranger automation flow



- Values become generated bootstrap configuration.
- Embedded Python reconciles Ranger state.
- Jobs and CronJobs handle bootstrap, usersync, local-user sync, and exception audits.
- The runtime target may be Ranger, LDAP or AD, or local Keycloak users.

# Where authorization actually happens

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| Situation                                                 | Ranger Admin? | Trino query authorization        |
|-----------------------------------------------------------|---------------|----------------------------------|
| Ranger disabled                                           | No            | Generated file-based Trino rules |
| Ranger enabled; ranger.trino=false                        | Yes           | Generated file-based Trino rules |
| Ranger enabled; ranger.trino=true; compatible Trino image | Yes           | Ranger plugin path               |

# Ranger templates are behavior-heavy

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## Where to read code

- `templates/ranger-automation.yaml` embeds reconciliation logic.
- `templates/_ranger-admin.tpl` owns bootstrap text templates.
- `templates/ranger-admin.yaml` renders runtime shell resources.
- `templates/ranger-browser-proxy.yaml` protects browser access.

**templates/ranger-automation.yaml embeds reconciliation logic.**



## Guided checkpoint: diagnose 'Ranger is enabled but Trino still uses file rules'

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Check `global.authorization.ranger.enabled`

Check `global.authorization.ranger.trino.enabled`

Check the Trino image/plugin compatibility expectation

Inspect rendered access-control config for coordinator

### Checkpoint questions

- Can the developer explain why Ranger Admin may be present but not query-time enforcing?
- Can they find the template path that renders Trino access control?

# Session 06 landing

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## What the inheriting team should now be able to do

- They know the governance values that matter.
- They understand why validation is strict for governed data.
- They can explain Ranger automation at a high level.
- They can distinguish Ranger Admin from Trino enforcement.

## Leave-behind

Use the PPTX notes as a presenter script and source-map.md as the reference trail.